

ECO 351: HW2

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1.

(a) The model is given as:

$$U_{ij}^* = x'_{ij}\beta + \varepsilon_{ij},$$

$$y_i = k, \text{ if } U_{ik}^* > U_{ij}^*, \forall j \neq k$$

Since the error terms follow the Gumbel distribution, the probability distribution for ε_{ij} is given as

$$f(\varepsilon_{ij}) = \exp[-\varepsilon_{ij} - \exp(-\varepsilon_{ij})]$$

The CDF for ε_{ij} is

$$F(\varepsilon_{ij}) = \exp[-\exp(-\varepsilon_{ij})]$$

Here, let $V_{ij} = x'_{ij}\beta$. According to the setup of our model,

$$\begin{aligned} Pr(y_i = j) &= Pr(U_{ij}^* > U_{ik}^*, \forall k \neq j) \\ &= Pr(V_{ij} + \varepsilon_{ij} > V_{ik} + \varepsilon_{ik}) \\ &= Pr(V_{ij} - V_{ik} + \varepsilon_{ij} > \varepsilon_{ik}) \end{aligned}$$

Imposing the CDF of the Gumbel distribution and treating ε_{ij} as a conditioning variable, we have

$$Pr(U_{ij}^* > U_{ik}^* | \varepsilon_{ij}) = F(V_{ij} - V_{ik} + \varepsilon_{ij})$$

Since all x'_{ik} are independent,

$$Pr(U_{ij}^* > U_{ik}^*, \forall k \neq j | \varepsilon_{ij}) = \prod_{k \neq j} F(V_{ij} - V_{ik} + \varepsilon_{ij})$$

Formally, the unconditional probability is written as

$$\begin{aligned} Pr(y_i = j) &= Pr(U_{ij}^* > U_{ik}^*, \forall k \neq j) \\ &= \int Pr(U_{ij}^* > U_{ik}^*, \forall k \neq j | \varepsilon_{ij}) \cdot f(\varepsilon_{ij}) d\varepsilon_{ij} \\ &= \int_{-\infty}^{\infty} \prod_{k \neq j} F(V_{ij} - V_{ik} + \varepsilon_{ij}) \cdot f(\varepsilon_{ij}) d\varepsilon_{ij} \end{aligned}$$

Imposing the parameterization,

$$Pr(y_i = j) = \int_{-\infty}^{\infty} \prod_{k \neq j} \exp(-\exp(-(V_{ij} - V_{ik} + \varepsilon_{ij}))) \cdot \exp(-\varepsilon_{ij}) \cdot \exp(-\exp(-\varepsilon_{ij})) d\varepsilon_{ij}$$

We can write $\exp(-\exp(-\varepsilon_{ij})) = \exp(-\exp(-(V_{ij} - V_{ik} + \varepsilon_{ij})))$. We can thus simplify as

$$\begin{aligned} Pr(y_i = j) &= \int_{-\infty}^{\infty} \prod_k \exp(-\exp(-(V_{ij} - V_{ik} + \varepsilon_{ij}))) \cdot \exp(-\varepsilon_{ij}) d\varepsilon_{ij} \\ &= \int_{-\infty}^{\infty} \exp\left(-\sum_k \exp(-(V_{ij} - V_{ik} + \varepsilon_{ij}))\right) \cdot \exp(-\varepsilon_{ij}) d\varepsilon_{ij} \\ &= \int_{-\infty}^{\infty} \exp\left(-\exp(-\varepsilon_{ij}) \sum_k \exp(-(V_{ij} - V_{ik}))\right) \cdot \exp(-\varepsilon_{ij}) d\varepsilon_{ij} \end{aligned}$$

$$\begin{aligned} \text{let } t &= -\exp(-\varepsilon_{ij}) \\ dt &= \exp(-\varepsilon_{ij}) d\varepsilon_{ij} \end{aligned}$$

$$\begin{aligned} \therefore Pr(y_i = j) &= \int_{-\infty}^0 \exp\left(t \sum_k \exp(-(V_{ij} - V_{ik}))\right) dt \\ &= \frac{\exp(t \sum_k \exp(-(V_{ij} - V_{ik})))}{\sum_k \exp(-(V_{ij} - V_{ik}))} \Big|_{-\infty}^0 \\ &= \frac{1}{\sum_k \exp(-V_{ij}) \exp(V_{ik})} \\ &= \frac{\exp(V_{ij})}{\sum_k \exp(V_{ik})} \end{aligned}$$

Hence, we have the final **choice probability**

$$\boxed{Pr(y_i = j) = \frac{\exp(x'_{ij}\beta)}{\sum_k \exp(x'_{ik}\beta)}}$$

The **Likelihood function** is written as:

$$\mathcal{L}(\beta) = \prod_{i=1}^n \left[\sum_{j=1}^J Pr(y_i = j) I(y_i = j) \right]$$

(b) Covariate effect of $x_{i1,2}$ on alternative 1 can be expressed as

$$\frac{\partial P_{i1}}{\partial x_{i1,2}} = \frac{\partial \left(\frac{\exp(\beta_{11} + \beta_2 x_{i1,2})}{\exp(\beta_{11} + \beta_2 x_{i1,2}) + \exp(\beta_{12} + \beta_2 x_{i2,2}) + \exp(\beta_{13} + \beta_2 x_{i3,2})} \right)}{\partial x_{i1,2}}$$

Let the denominator of the P_{i1} term be denoted as *denom* for simplicity

$$\begin{aligned} &= \frac{[\beta_2 \exp(\beta_{11} + \beta_2 x_{i1,2})][denom] - \exp(\beta_{11} + \beta_2 x_{i1,2})[\beta_2 \exp(\beta_{11} + \beta_2 x_{i1,2})]}{(denom)^2} \\ &= \frac{\beta_2 \exp(\beta_{11} + \beta_2 x_{i1,2})[\exp(\beta_{12} + \beta_2 x_{i2,2}) + \exp(\beta_{13} + \beta_2 x_{i3,2})]}{(denom)(denom)} \end{aligned}$$

Separating the numerators appropriately, we get

$$\boxed{\frac{\partial P_{i1}}{\partial x_{i1,2}} = \beta_2 P_{i1} (1 - P_{i1})}$$

(c) The covariate effect of $x_{i1,2}$ on alternative 2 can be expressed as

$$\frac{\partial P_{i2}}{\partial x_{i1,2}} = \frac{\partial \left(\frac{\exp(\beta_{12} + \beta_2 x_{i2,2})}{\exp(\beta_{11} + \beta_2 x_{i1,2}) + \exp(\beta_{12} + \beta_2 x_{i2,2}) + \exp(\beta_{13} + \beta_2 x_{i3,2})} \right)}{\partial x_{i1,2}}$$

Let the denominator of P_{i2} and P_{i3} be denoted as *denom* for simplicity

$$= \frac{(\text{denom})(0) - \exp(\beta_{12} + \beta_2 x_{i2,2}) [\beta_2 \exp(\beta_{11} + \beta_2 x_{i1,2})]}{(\text{denom})(\text{denom})}$$

Rearranging the terms gives us

$$\boxed{\frac{\partial P_{i2}}{\partial x_{i1,2}} = -\beta_2 P_{i1} P_{i2}}$$

Similarly, the covariate effect of $x_{i1,2}$ on alternative 3 can be expressed as

$$\begin{aligned} \frac{\partial P_{i3}}{\partial x_{i1,2}} &= \frac{\partial \left(\frac{\exp(\beta_{13} + \beta_2 x_{i3,2})}{\exp(\beta_{11} + \beta_2 x_{i1,2}) + \exp(\beta_{12} + \beta_2 x_{i2,2}) + \exp(\beta_{13} + \beta_2 x_{i3,2})} \right)}{\partial x_{i1,2}} \\ &= \frac{(\text{denom})(0) - \exp(\beta_{13} + \beta_2 x_{i3,2}) [\beta_2 \exp(\beta_{11} + \beta_2 x_{i1,2})]}{(\text{denom})(\text{denom})} \end{aligned}$$

Rearranging the terms gives us

$$\boxed{\frac{\partial P_{i3}}{\partial x_{i1,2}} = -\beta_2 P_{i1} P_{i3}}$$

2.

(a) We have

$$Pr(y_i = j) = \frac{\exp(x'_{ij}\beta)}{\sum_k \exp(x'_{ik}\beta)}$$

Hence, for our model:

$$P_{ij} = \frac{\exp(\text{price}'_{ij}\beta)}{\sum_{k=1}^3 \exp(\text{price}'_{ik}\beta)}$$

In the given situation, $P(\text{outcome}) = P_{11}P_{22}P_{33}$, since the individual choices are independent.

$$P(\text{outcome}) = \frac{\exp(\beta_{11} + \beta_2 \text{price}_{11})}{\exp(\beta_{11} + \beta_2 \text{price}_{11}) + \exp(\beta_{12} + \beta_2 \text{price}_{12}) + \exp(\beta_2 \text{price}_{13})} \times \frac{\exp(\beta_{12} + \beta_2 \text{price}_{22})}{\exp(\beta_{11} + \beta_2 \text{price}_{21}) + \exp(\beta_{12} + \beta_2 \text{price}_{22}) + \exp(\beta_2 \text{price}_{23})} \times \frac{\exp(\beta_2 \text{price}_{33})}{\exp(\beta_{11} + \beta_2 \text{price}_{31}) + \exp(\beta_{12} + \beta_2 \text{price}_{32}) + \exp(\beta_2 \text{price}_{33})}$$

(b)

	Estimates	SE	tStat
β_{11}	0.283	0.062	4.539
β_{12}	0.104	0.062	1.662
β_2	-2.296	0.138	-16.681

(c) The predicted choice probabilities for $p_{i1} = 1.00$, $p_{i2} = 1.25$ and $p_{i3} = 1.10$ are: $P(\text{Pepsi}) = 0.483$, $P(7\text{-Up}) = 0.227$ and $P(\text{Coke}) = 0.289$

(d) When price of Pepsi increases by \$0.10:

Change in $P(\text{pepsi}) = -0.0573$

Change in $P(7\text{-up}) = 0.0252$

Change in $P(\text{Coke}) = 0.0321$

When price of 7-Up increases by \$0.10:

Change in $P(\text{pepsi}) = 0.0252$

Change in $P(7\text{-up}) = -0.0404$

Change in $P(\text{Coke}) = 0.0151$

When price of Coke increases by \$0.10:

Change in $P(\text{pepsi}) = 0.0321$

Change in $P(7\text{-up}) = 0.0151$

Change in $P(\text{Coke}) = -0.0472$

3.

(a)

		Mean	Median	Std dev	Max	Min
East	"FMD1998"	1.1923	0	2.4003	9	0
	"Cattle"	170080	121675	1.481e+05	622650	43760
West	"FMD1998"	1.0750	0	1.6546	5	0
	"Cattle"	157850	135025	8.2619e+04	392840	39330

We observe that the values for Cattle in both regions are much much larger than those for FMD1998. We also observe that all values in FMD1998 are non-negative and we want to predict the counts of foot and mouth diseases in cattle which is why we try to fit the

Poisson regression model which is a Count Data Model.

We also observe that the mean of cattle having FMD is higher in the East than in the West. The maximum number of FMD incidences reported in a province of East is 9 and West is 5.

	Estimates	SE	tStat	95% CI
(b) β_1	-11.254	2.280	-4.936	[-15.8105, -6.6979]
β_2	0.947	0.188	5.029	[0.5706, 1.3231]
β_3	0.041	0.239	0.173	[-0.4370, 0.5197]

- (c) From the table, we can say that a 1% increase in $\ln \text{Cattle}$ increases expected FMD by 0.947%. The absolute value of the critical t-value for 63 degrees of freedom and 95% confidence interval is nearly 2. Here, the absolute value of tStat for $\beta_2 = 5.029 > 2$ which implies that the result is statistically significant and we can reject the null hypothesis of $\ln \text{Cattle}$ having no effect on FMD incidents.

For β_3 , being in an Eastern province multiplies the expected FMD by e^{β_3} as opposed to being in a Western province. However, the tStat for $\beta_3 = 0.173 < 2$ which implies that we cannot reject the null hypothesis of being in Eastern Turkey having no effect on FMD instances. β_3 is not statistically significant.

- (d) Expected number of FMD in Eastern Turkey (when Cattle= 170080)= 1.210.
(e) $P(\text{FMD} \leq 2)$ in Eastern Turkey= 0.8773
(f) Expected number of FMD in Western Turkey (when Cattle= 157850)= 1.082
(g) $P(\text{FMD} \leq 2)$ in Western Turkey= 0.9040

4.

- (a) We can write the ZIP model as:

$$y = \begin{cases} 0 & \text{prob. } \pi \\ \text{Poisson}(\lambda) & \text{prob. } 1 - \pi \end{cases} \quad 0 < \pi < 1 \text{ and } 0 < e^{-\lambda} < 1$$

There are two ways to get a zero outcome:

- (i) The draw came from the structural-zero component: probability π
(ii) The draw came from the Poisson component ($1 - \pi$ prob) and the Poisson produced a zero. Probability($P_{\text{pois}}(0)$)= $e^{-\lambda}$

$$Pr_{ZIP}(y = 0) = \pi + (1 - \pi)e^{-\lambda}$$

However, when we take the Poisson alone

$$Pr_{\text{pois}}(y = 0) = e^{-\lambda}$$

If we take the difference of the two,

$$\begin{aligned} Pr_{ZIP}(y = 0) - Pr_{pois}(y = 0) &= \pi + (1 - \pi)e^{-\lambda} - e^{-\lambda} \\ &= \pi(1 - e^{-\lambda}) \end{aligned}$$

We have $1 - e^{-\lambda} > 0$

$$Pr_{ZIP}(y = 0) > Pr_{pois}(y = 0)$$

Hence, the overall probability of observing a zero outcome for this mixture distribution is larger than the probability of observing zero flowing from a Poisson distribution alone.

- (b) If $y_i = 0 : f(y_i = 0) = \pi + (1 - \pi)e^{-\lambda}$
 If $y_i = k$ with $k \geq 1 : f(y_i = k) = (1 - \pi) \frac{e^{-\lambda} \lambda^k}{k!}$
 We use an Indicator variable $I\{y_i = 0\}$
 For just a single observation,

$$l(\lambda, \pi; y_i) = [\pi + (1 - \pi)e^{-\lambda}]^{I\{y_i=0\}} [(1 - \pi) \frac{e^{-\lambda} \lambda^{y_i}}{y_i!}]^{I\{y_i>0\}}$$

For **n observations**, we multiply all the likelihoods for individual observations.

$$\begin{aligned} \mathcal{L}(\lambda, \pi; \{y_i\}) &= \prod_{i:y_i=0}^n [\pi + (1 - \pi)e^{-\lambda}]^{I\{y_i=0\}} \cdot \prod_{i:y_i>0}^n [(1 - \pi) \frac{e^{-\lambda} \lambda^{y_i}}{y_i!}]^{I\{y_i>0\}} \\ \mathcal{L}(\lambda, \pi; \{y_i\}) &= \prod_{i:y_i=0}^n [\pi_i + (1 - \pi_i)e^{-\lambda_i}] \cdot \prod_{i:y_i>0}^n [(1 - \pi_i) \frac{e^{-\lambda_i} \lambda_i^{y_i}}{y_i!}] \end{aligned}$$

Note that π_i can be a link function to a probability distribution.

Log-likelihood:

$$L(\theta) = \sum_{i:y_i=0} \log[\pi_i + (1 - \pi_i)e^{-\lambda_i}] + \sum_{i:y_i>0} [\log(1 - \pi_i) - \lambda_i + y_i \log \lambda_i - \log(y_i!)]$$

where $\theta \rightarrow$ parameters governing $\{\pi_i, \lambda_i\}$